Steering a nominal diffusion model to synthesize spacecraft anomalies

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Abstract

Seven years of telemetry on six channels of one spacecraft contain 22 labelled anomalies: enough to evaluate a detector, and too few to train one or to train a generator on the anomalies themselves. We train a diffusion model on nominal operation only and steer it at sampling time. Each sample starts from a partly noised nominal window, and at every denoising step the predicted clean window is edited by two terms. A shell term moves a soft encoder distance to nominal reference windows toward its 99th percentile on nominal data. On spikes and level shifts, an exact linear projection sets a weighted difference of means to an amplitude resampled from real examples, at a sampled position. On the development split of ESA-ADB Mission 1, channels 41–46, the generated spikes match the real ones in height (0.838 against 0.831 of the channel span) and width (99% occupy a single bin), and the generated level shifts carry the real step amplitude. Added to a small detector, the windows raise event recall when real anomalies are scarce: by +0.20 at 32 real training windows, 95% interval $[+0.07, +0.32]$, with intervals above zero also at 8 and 64 windows. With every development anomaly available, the gain of +0.09 is unresolved. Two baselines that pull each window toward the embedding of a real anomaly score higher on feature-space realism or gain more at the smallest budgets; they plant spikes at about half the real height and inherit the position of the labelled windows. Published editors applied to the same donors score at the level of the unedited donors. The sealed test split is unopened and reserved for one pre-registered run.

This draft reports the development split only. Section 6.4 is a placeholder for a single pre-registered run on the sealed test events.

1 Introduction

Anomaly detection in spacecraft telemetry is limited by labels. A channel can be recorded for years and still hold a few dozen annotated events, of several shapes, many of which occur once [14, 11]. A detector trained on those events learns the ones it has seen. A generator trained on them has almost nothing to fit.

Nominal operation, in contrast, is abundant. We learn it with a diffusion model and request anomalies only at sampling time. Once the denoiser is trained, a sample can be edited by changing the predicted clean window before the chain re-noises it [10, 21, 18]. Classifier guidance and its successors use this mechanism for images and for time series [7, 5, 4, 1]. Each of them needs a target written as a function of the window, and "be anomalous" becomes usable only once it is written that way.

We write such targets for the four anomaly kinds that occur on our channels of ESA-ADB [14, 2]. An isolated spike and a level shift are linear contrasts: a weighted difference of means with a known amplitude. The smallest edit that meets a linear constraint is a projection, which is exact in one step and independent of the scale of the weight vector. The other two kinds are long

subsequences, one outside the channel's usual range and one outside the range expected from its context. They receive a shell term: a soft distance to nominal reference windows in a frozen encoder, moved toward the 99th percentile of the same distance on nominal windows. The shell asks for windows that are unusual among nominal data, and it runs on every window.

We evaluate the samples in three ways, all held out by event on the development events. The first is the realism and diversity protocol of GenIAS [26] on a fixed twelve-dimensional feature of each window. The second measures the shape of the generated spikes and level shifts directly. The third asks whether the samples help a small convolutional detector, scored by event recall at a fixed false-alarm rate as the number of real training anomalies varies. The 91 test events stay sealed for a single pre-registered run.

The contributions are:

- per-kind targets for a nominal diffusion model: an exact contrast projection for spikes and level shifts, with explicit weight vectors, and a shell quantile for every window (Sections 4.3 and 4.4);
- a sampler for a causal nominal denoiser: SDEdit from nominal donors, a noise prediction blended from forward and time-reversed passes, edits on the clean-window prediction, one self-recurrence per step, and an SDEdit polish for level shifts (Sections 4.1 and 4.2, Algorithm 1);
- an evaluation on ESA-ADB Mission 1 that pairs feature-space realism, direct morphology measurements and detector utility at matched real-data budgets, with intervals from a bootstrap over events (Section 5).

The generated spikes match the real height and width, and the level shifts match the real step amplitude. Added to k real windows, the gallery raises event recall with an interval above zero at $k = 8$, 32 and 64; at the full budget the gain is unresolved. Two prototype baselines, which pull each window toward the embedding of a real anomaly, score higher on feature-space realism or gain more at the smallest budgets, with spikes at half the real height and positions copied from the labelled windows. We report both sides of that trade. The recipe in this paper is frozen; changes to the backbone are future work.

2 Related work

Spacecraft telemetry is a poor match for detectors tuned on dense, repeated labels. Events are rare, annotations cover an interval, and a false alarm has an operational cost [11, 14]. ESA-ADB is the public form of that problem, with a shared catalogue of anomaly types [14, 2, 15]. We use that catalogue as given.

When labels are scarce, the usual synthetic anomaly is drawn by hand: a step, a pulse, a ramp, or a change of scale [25, 3]. CutAddPaste pastes a real patch into another window [24]. GenIAS trains a variational generator to occupy regions of a feature space where real anomalies sit, and defines the realism and diversity scores we report [26]. All three edit an existing window, and so do we. Diffusion models that generate or impute whole series from noise [22, 13] are outside this comparison.

Classifier guidance adds the gradient of a label to the diffusion score [7]. Later work aims the same mechanism at low-density regions and minority samples [20, 23] and at explicit time-series constraints [5], and applies the correction to the clean-window prediction [4, 1]. Our shell term belongs to the low-density family: it targets a fixed quantile of a nominal energy, which places the sample at a controlled distance from the nominal set. SDEdit starts from a partly noised observation so that its coarse structure survives [18]; RePaint confines the edit to a mask [17]. Our sampler is an SDEdit chain with one self-recurrence per step [1] and a causal S4 denoiser of the kind used by TSDiff [9, 13].

Table 1: Anomaly windows on the development split, channels 41–46. Several windows belong to one event, so the event count is the relevant sample size for every comparison below.

Kind	Windows	Events
Point / Global	18	3
Level shift	6	1
Global subsequence	636	17
Local subsequence	12	2
All anomalies	672	22

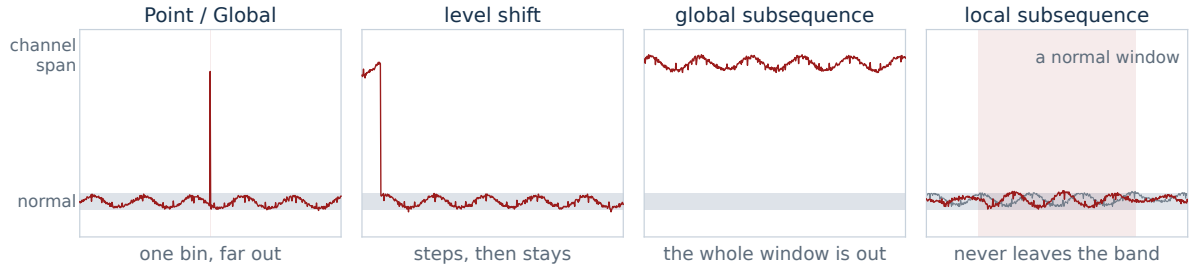


Figure 1: One real anomalous window of each kind, channel 41, in channel-span units. The grey band is the 1st to 99th percentile of nominal windows on the same channel. Where the label covers only part of the window, the labelled bins are shaded. The local subsequence is drawn over a nominal window (grey line): it stays inside the band and differs in its pattern.

A prototype attraction toward a real anomaly also attracts the sample toward that anomaly's position. On ESA-ADB, windows of short events are cut with the event in the middle, so a prototype reproduces the middle. A contrast specifies shape and amplitude and leaves the position to be sampled.

3 Data

We use ESA-ADB Mission 1, channels 41–46 [14]. Each channel is a regular series of 30 s bins. The development period runs from 1 January 2000 to 31 December 2006, 7,364,160 bins per channel; the test period starts on 1 January 2007. Windows are $W = 512$ bins, about 4.3 hours.

The mission catalogue has 200 annotated events. We keep the benchmark split: the 109 events of the development period are used below, and the 91 events of the test period form a sealed test. The data loader refuses to read test-period telemetry. Of the development events, 22 anomalies and 27 rare but valid operations have windows on channels 41–46; the others produce none. Every one of the 22 anomalies is annotated on all six channels. An event shorter than W gives one window centred on its midpoint, and a longer event gives up to eight evenly spaced windows inside it. This yields 112 anomaly windows per channel, 672 in total, and 943 rare windows. Table 1 lists the anomaly kinds, and Figure 1 shows one real window of each.

The imbalance matters. A method that only ever produced a plausible global subsequence would already match 636 of the 672 windows and 17 of the 22 events. The level shift is a single event, so anything we report about level shifts is a statement about that event.

Nominal windows are drawn at random starts in everyday operation, 4,096 per channel, at least 16 bins away from any labelled span. These 24,576 windows and the 943 rare windows form the training pool of the diffusion model, which never sees an anomaly. A gallery of 1,536 nominal donors, 256 per channel drawn from the nominal windows, is the starting point of every generated sample, for our method and for every baseline.

The development anomalies are split into three folds by event: 8, 7 and 7 events, with 270,

276 and 126 windows. Every comparison below holds one fold out, so the training folds hold 402, 396 or 546 anomaly windows. A fold's held-out events never set that fold's contrast amplitudes, with one exception for the level shift, discussed in Section 4.4.

4 Method

Figure 2 shows how one gallery is made, and Algorithm 1 gives the chain for a single window. A diffusion model is trained on nominal windows (Section 4.1). Generation runs it from partly noised donors (Section 4.2). At every step the predicted clean window is edited by a shell term on every window (Section 4.3) and by a contrast term on spikes and level shifts (Section 4.4). The windows of the subsequence pool are named after sampling (Section 4.5).

4.1 Nominal diffusion model

One scaling. For each channel c, a minimum $\underline{x}_c$ and a maximum $\overline{x}_c$ are fit once on the training pool, and

$$m(x) = \frac{x - \underline{x}_c}{\overline{x}_c - \underline{x}_c} \tag{1}$$

maps the channel onto $[0, 1]$. We call these channel-span units; every amplitude in this paper is given in them. The map is the only scaling in the recipe. A donor is mapped at the start of sampling, the noisy state stays in these units throughout the chain, and m^{-1} is applied once to the finished window, which returns it to panel units. Three requirements fix this choice. The diffusion noise is standard normal, so the clean signal must be of order one for the schedule's signal-to-noise ratios to hold [10]. The contrast targets of Section 4.4 are measured on real windows after the same map, so a target means the same step on every donor of a channel; a per-window normalization would rescale it by each window's own range. And the steering encoder was trained in panel units: it reads $m^{-1}(\hat{x}_0)$, and because m is affine this multiplies the shell gradient by a per-channel constant that the normalization of the shell step removes.

Forward process and denoiser. The forward process is $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$, $\bar{\alpha}_t = \prod_{s \le t}(1 - \beta_s)$, $T = 200$ steps, and β linear from 10^{-4} to $2 \cdot 10^{-2}$ [10]. The network $\varepsilon_\theta(x_t, t, c)$ predicts the added noise, and the clean-window prediction is

$$\hat{x}_0(x_t) = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\, \varepsilon(x_t)}{\sqrt{\bar{\alpha}_t}}. \tag{2}$$

The network follows the S4 backbone of TSDiff [13, 9]. A pointwise convolution lifts the window to 64 channels, and a learned embedding of the channel index c is added. Six residual blocks follow. Each applies layer normalization, a causal diagonal S4 layer with state size 64, a FiLM scale and shift computed from a 128-dimensional sinusoidal embedding of t passed through a two-layer perceptron, and a pointwise feed-forward layer with a GELU, and adds the block's input. A final layer normalization and pointwise convolution return one channel. The network has 315,841 parameters. It is trained for 20,000 steps of noise-prediction loss, batch size 128, learning rate $2 \cdot 10^{-4}$, on the 25,519 windows of the training pool, of which 10% are held out for validation.

Reading the window both ways. The S4 layers are causal, so the network denoises bin i from bins 0 to i. At sampling time we evaluate it on the window and on its time reversal and blend the two noise predictions with a linear ramp,

$$\varepsilon(x)_i = r_i\, \varepsilon_\theta(x)_i + (1 - r_i)\, \big(\varepsilon_\theta(\mathrm{flip}\, x)\big)_{\mathrm{flip},\, i}, \qquad r_i = \frac{i}{W - 1}. \tag{3}$$

The forward pass carries the most weight at the end of the window, where it has seen the most context, and the reversed pass at the start. Both passes use the same weights, and the blend assumes that the nominal process is approximately reversible in time. Every network evaluation below means this blend of two passes.

4.2 From donor to window

Donors and kinds. Each channel's 256 donors are split into four groups of 64: spikes, level shifts, and two groups that together form the subsequence pool of 128 windows. A spike or level shift is requested before sampling and keeps its name. The subsequence pool is named after sampling (Section 4.5).

Starting point. A donor u is mapped by (1) and noised to

$$t_0 = \operatorname{round}\big(0.2 \cdot (T-1)\big) = 40, \qquad x_{t_0} = \sqrt{\bar\alpha_{40}}\, m(u) + \sqrt{1-\bar\alpha_{40}}\, \xi, \tag{4}$$

where ξ is standard normal and $\bar\alpha_{40} = 0.917$. The noise is then 0.30 of the clean signal's scale, so the donor's slow structure survives and its fine texture is drawn again [18].

Time grid and step. The chain takes 50 DDIM steps [21] on the grid $t_n = \lfloor 40\,(1 - n/49) \rfloor$, $n = 0, \ldots, 49$. Rounding down gives 41 distinct indices; nine of them occur twice in a row, and a step between two equal indices keeps t fixed. From x_t at $t = t_n$ to $t' = t_{n+1}$, the step is the ordinary DDIM update applied to an edited clean window,

$$x_{t'} = \sqrt{\bar\alpha_{t'}}\,\big(\hat x_0 - E_t\big) + \sqrt{1-\bar\alpha_{t'}}\;\varepsilon(x_t). \tag{5}$$

The edit $E_t \in \mathbb{R}^W$ is the shell term E_{shell} of Section 4.3 on every window, plus the contrast term E_{con} of Section 4.4 on spikes and level shifts. Both are functions of $\hat x_0$ alone. The denoiser is evaluated without gradients, and the shell gradient passes through the encoder only, which is the clean-window form of guidance [1].

Self-recurrence and the last step. Whenever $t' < t$, the new state is re-noised back to t,

$$x_t \leftarrow \sqrt{\bar\alpha_t/\bar\alpha_{t'}}\; x_{t'} + \sqrt{1-\bar\alpha_t/\bar\alpha_{t'}}\;\xi, \tag{6}$$

and the step is run once more [1]. A step between equal indices has no noise interval to travel back over and runs once. At $n = 49$ the chain returns $\hat x_0 - E_0/4$. No later step remains to blend this last edit into the window, so it enters at a quarter of its strength. In total the chain makes 90 network evaluations: 40 steps run twice, nine run once, and the last step.

Shell weights. The shell term is scaled along the chain by

$$\omega_t = \frac{\sigma_t}{\frac{1}{50}\sum_{n=0}^{49}\sigma_{t_n}}, \qquad \sigma_t = \sqrt{\frac{1-\bar\alpha_t}{\bar\alpha_t}}, \tag{7}$$

which falls from 2.0 at $t = 40$ to 0.067 at $t = 0$. The shell edit is large while the chain still has noise to remove and fades as the window settles. The contrast term keeps weight one throughout: it is an exact projection, and its size is already set by its residual.

Polish for level shifts. After the chain, a level shift is re-noised to $t = \operatorname{round}(0.35 \cdot (T-1)) = 70$, where $\bar\alpha_{70} = 0.774$, denoised by ten unguided DDIM steps, and projected once more by E_{con}. This SDEdit pass [18] lets the nominal denoiser rebuild the texture around the step. The denoiser also shrinks the step, and the final projection restores its amplitude exactly.

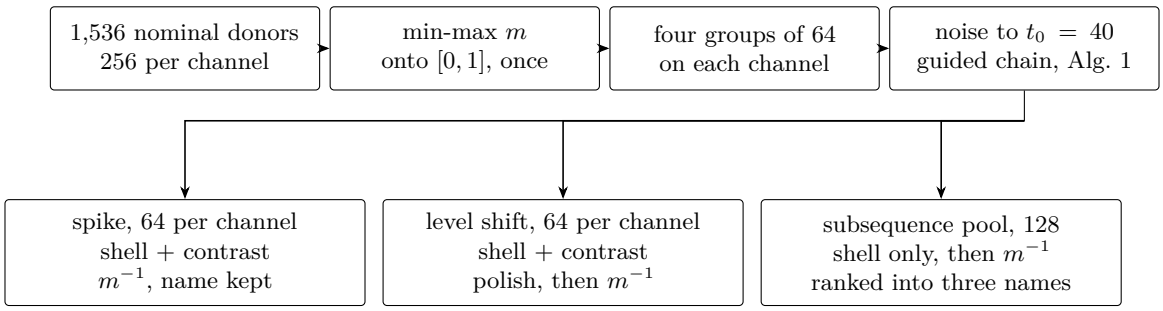


Figure 2: One generated gallery. The min-max map is applied once, at the start, and inverted once, at the end. Spikes and level shifts keep the kind they were assigned; the subsequence pool is named after sampling.

Algorithm 1 One steered window. ξ is fresh standard normal noise at each use, $\varepsilon(x, t)$ the blended prediction (3), and $\hat{x}_0$ follows from (2). E_{con} is zero on the subsequence pool.

Require: donor u on channel c; for a spike or level shift, weight w and target δ

$x \leftarrow \sqrt{\bar{\alpha}_{40}}\, m(u) + \sqrt{1 - \bar{\alpha}_{40}}\, \xi$
for $n = 0, \ldots, 48$ **do**
  $t \leftarrow t_n$, $t' \leftarrow t_{n+1}$, $P \leftarrow 2$ if $t' < t$, else 1
  for $p = 1, \ldots, P$ **do**
    $\varepsilon \leftarrow \varepsilon(x, t)$, $\hat{x}_0 \leftarrow \hat{x}_0(x)$
    $E \leftarrow E_{\text{shell}}(\hat{x}_0, t) + E_{\text{con}}(\hat{x}_0)$
    $x \leftarrow \sqrt{\bar{\alpha}_{t'}}\,(\hat{x}_0 - E) + \sqrt{1 - \bar{\alpha}_{t'}}\,\varepsilon$
    if $p < P$ **then**
      $x \leftarrow \sqrt{\bar{\alpha}_t / \bar{\alpha}_{t'}}\; x + \sqrt{1 - \bar{\alpha}_t / \bar{\alpha}_{t'}}\; \xi$ ▷ self-recurrence
    end if
  end for
end for
$\hat{x}_0 \leftarrow \hat{x}_0(x)$ at $t = 0$; $x \leftarrow \hat{x}_0 - \frac{1}{4}\big(E_{\text{shell}}(\hat{x}_0, 0) + E_{\text{con}}(\hat{x}_0)\big)$
if level shift **then**
  $x \leftarrow \sqrt{\bar{\alpha}_{70}}\, x + \sqrt{1 - \bar{\alpha}_{70}}\, \xi$; ten unguided DDIM steps from $t = 70$ to 0
  $x \leftarrow x - E_{\text{con}}(x)$ ▷ polish
end if
return $m^{-1}(x)$

4.3 Shell term

The shell term asks for a window that is unusual among nominal windows by a controlled amount. It works in the embedding space of a frozen encoder e, described below, with $z = e\big(m^{-1}(\hat{x}_0)\big) \in \mathbb{R}^{512}$.

Energy. Let $\zeta_1, \ldots, \zeta_N$ be the embeddings of $N = 512$ nominal reference windows. The soft energy of an embedding z is

$$h(z) = -\tau \log \sum_{j=1}^{N} \exp\big(-\|z - \zeta_j\|^2 / \tau\big), \tag{8}$$

a soft minimum of squared distance to the references. As $\tau \to 0$ it becomes the squared distance to the nearest reference, and with a single reference the two factors of τ cancel and $h(z) = \|z - \zeta\|^2$. The factor inside the exponential decides which references still contribute; the factor in front returns h to units of squared distance. A large h means far from the nominal references. The sum also contributes $-\tau \log N$, so h can be negative.

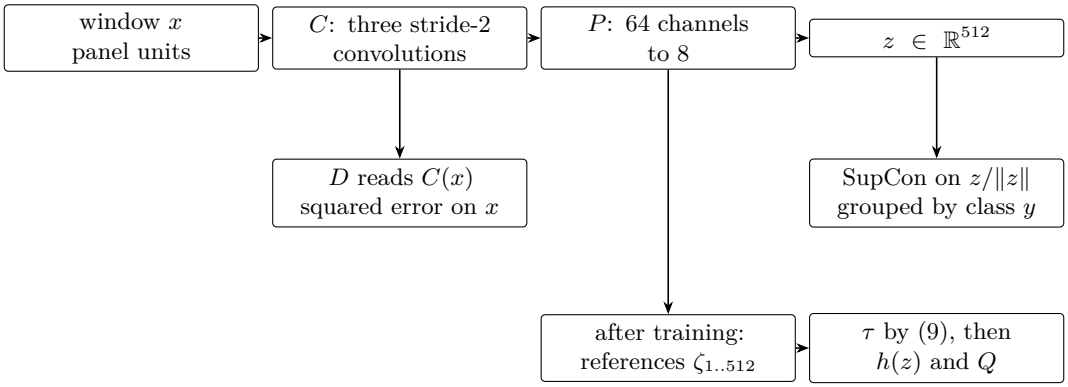


Figure 3: The steering encoder. The top row and the two losses are training; D reads the 64-channel map, and the class y enters only the contrastive loss. After the weights are frozen, donor embeddings supply the references, τ and Q of the shell energy.

Temperature and target. The temperature is set from the reference embeddings themselves,

$$\tau = \sqrt{\text{median}_{i \neq j} \, \|\zeta_i - \zeta_j\|^2}, \tag{9}$$

with a floor of 10^{-3}. The embeddings are unnormalized, and the square root of the median squared distance puts the kernel on their scale. The target Q is the 0.99 quantile of h over 256 further donors. A nominal window exceeds Q one time in a hundred, so Q marks the outer edge of ordinary operation. The references and the 256 windows are drawn independently from the 1,536 donors, with a fixed seed, when sampling starts. The encoder checkpoint stores weights only; τ and Q are computed at that point.

Step. The shell loss is $(h - Q)^2$, and the edit is its gradient with respect to the clean window, normalized over the window,

$$E_{\text{shell}} = \lambda \, \omega_t \, \frac{g}{\|g\|}, \qquad g = \nabla_{\hat{x}_0} \big(h(z) - Q\big)^2, \qquad \lambda = 0.3. \tag{10}$$

With $J = \partial z / \partial \hat{x}_0$, the chain rule gives

$$g = 2\,(h - Q)\, J^\top \nabla_z h, \qquad \nabla_z h = 2 \sum_{j=1}^{N} \pi_j \, (z - \zeta_j), \qquad \pi_j = \frac{\exp(-\|z - \zeta_j\|^2/\tau)}{\sum_l \exp(-\|z - \zeta_l\|^2/\tau)}. \tag{11}$$

After normalization, $E_{\text{shell}} = \lambda \, \omega_t \, \text{sign}(h - Q)\, J^\top \nabla_z h / \|J^\top \nabla_z h\|$. The step has length $\lambda \, \omega_t$ on every window, and h and Q decide only its sign. The leading τ of (8) cancels from $\nabla_z h$. The inner τ sets the weights π_j and with them the direction: a small τ points the step away from the nearest reference, a large τ away from the references' common centre. Most donors start below Q, so subtracting E_{shell} in (5) raises h toward Q, and a window that overshoots Q is pulled back. The term runs on every window and is the only term on the subsequence pool.

Encoder. The encoder keeps the time axis. Its input is the clean window in panel units, the space it was trained in. Three stride-2 convolutions C ($1 \to 32$, kernel 7; $32 \to 64$, kernel 5; $64 \to 64$, kernel 5), each followed by a SiLU, reduce 512 bins to 64. A pointwise convolution P maps the 64 channels to 8, and $z \in \mathbb{R}^{512}$ is that 8×64 map flattened; C and P have 31,624 parameters. A decoder D of three stride-2 transposed convolutions, kernel 4 ($64 \to 32 \to 32 \to 1$ channels), reads the 64-channel map $C(x)$ and is used only in training. Figure 3 shows the two paths.

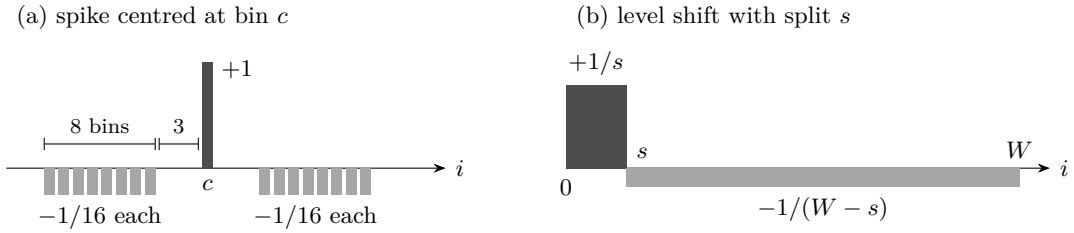


Figure 4: The two weight vectors, heights not to scale. (a) A spike: weight +1 on bin c and $-1/16$ on eight bins on each side, after a gap of three. (b) A level shift: the mean of the head minus the mean of the tail. Both vectors sum to zero. With probability 1/2 the level-shift vector is reversed in time.

Training minimizes a reconstruction loss and a supervised contrastive loss [12] with equal weight,

$$\mathcal{L} = \frac{1}{W}\|D(C(x)) - x\|^2 + \mathrm{SupCon}(v, y), \qquad v = z/\|z\|. \tag{12}$$

The class y is everyday nominal, rare-but-valid, or anomaly. With positives $\mathcal{P}(i) = \{p \neq i : y_p = y_i\}$ in a batch and temperature $\kappa = 0.1$,

$$\mathrm{SupCon} = \mathrm{mean}_{i\,:\,\mathcal{P}(i)\neq\emptyset}\left(-\frac{1}{|\mathcal{P}(i)|}\sum_{p\in\mathcal{P}(i)} \log \frac{\exp(v_i \cdot v_p/\kappa)}{\sum_{l\neq i}\exp(v_i \cdot v_l/\kappa)}\right). \tag{13}$$

Training runs for 1,200 Adam steps at learning rate 10^{-3}, each batch holding 32 windows of each class. One checkpoint, trained with fold 0 held out, serves every fold, so its rare and anomaly windows are those of folds 1 and 2 (Section 7).

4.4 Contrast term

A spike and a level shift are each specified by a linear functional $d(x) = \langle w, x\rangle$ and a target amplitude δ. The weight $w \in \mathbb{R}^W$ sums to zero, so d ignores the window's overall level. Figure 4 shows both weight vectors.

Spike. The positive weight sits on one bin. With the centre at bin c,

$$w_c = 1, \qquad w_i = -\frac{1}{16} \text{ for } i \in [c-11, c-4] \cup [c+4, c+11], \tag{14}$$

and $w_i = 0$ on the three bins to either side of c and everywhere else. Then $d(x)$ is x_c minus the mean of sixteen flank bins. The three-bin gap keeps the spike's own rise and fall out of the baseline estimate. The centre c is drawn uniformly from bins 51 to 459, where both flanks always fit.

Level shift. The split s is drawn uniformly from bins 32 to 91, which is 6.25% to 18% of the window and brackets the split of the real event. Head and tail carry total weights +1 and −1,

$$w_i = \begin{cases} +1/s & 0 \le i < s, \\ -1/(W-s) & s \le i < W, \end{cases} \tag{15}$$

so $d(x)$ is the mean of the short head minus the mean of the long tail. With probability 1/2 the weight is reversed in time, which moves the short side to the end of the window and keeps the amplitude.

Projection. Among all edits E with $\langle w, \hat{x}_0 - E\rangle = \delta$, the one of smallest Euclidean norm is

$$E_{\text{con}} = \big(\langle w, \hat{x}_0\rangle - \delta\big)\,\frac{w}{\|w\|^2}. \tag{16}$$

It moves $\hat{x}_0$ along w only, it meets the constraint exactly, and it is unchanged when w is multiplied by a constant. The division by $\|w\|^2$ matters for the level shift, where $\|w\|^2 = 1/s + 1/(W-s)$ changes with the split. A window whose contrast already equals δ is left as it is. The projection is evaluated in channel-span units, like the shell edit, and added to it.

Amplitude and sign. The target δ is resampled from real windows of the same kind in the other folds. On each real window, after m, d is evaluated at every admissible position and the value of largest magnitude is kept with its sign. The sign of a draw is flipped only when $|\delta|$ exceeds the room between the donor and the edge of $[0,1]$ in the requested direction, measured on the bins of positive weight, and the opposite direction has more room. The magnitude is always kept. The level shift has one real event, in fold 0, so folds 1 and 2 borrow its amplitude, and fold 0 measures the amplitude on its own held-out event.

4.5 Subsequence pool and gallery

The 128 pool windows of a channel receive the shell edit and no weight vector. After sampling they are named from the finished window and from the donors on the same channel. Let b_- and b_+ be the 5th and 95th percentiles of the donor bins. The out-of-range score is

$$g_{\text{out}}(x) = \frac{1}{W}\sum_i \big(\max(0, x_i - b_+) + \max(0, b_- - x_i)\big). \tag{17}$$

The contextual score clips the window into $[b_-, b_+]$ and compares a stretch with the rest of the window. Stretches are 64 bins long and start every 16 bins. Writing $\tilde{x}$ for the clipped window and S for a stretch,

$$g_{\text{ctx}}(x) = \max_S \big|\text{mean}_S(\tilde{x}) - \text{mean}_{\text{outside } S}(\tilde{x})\big|. \tag{18}$$

The top third of the pool by g_{out} is named global subsequence. Of the rest, the top third by g_{ctx} is named local subsequence, and the remainder is named subsequence, without a local or global claim. Since 128 is not divisible by three, the two spare windows rotate with the channel, and each of the three names totals 256 over the six channels. A gallery therefore holds 384 spikes, 384 level shifts and 256 windows of each subsequence name, a balance that differs from the real counts in Table 1.

The pool receives the shell and nothing else. A prototype pull toward a real anomaly embedding would carry the real window's layout with it (Section 2). The final recipe also has no spatial mask. An earlier variant replaced every bin outside a mask around the requested anomaly with the noised donor at each step, as in RePaint [17], which yields a bin-level label. The final recipe lets the chain rewrite the whole window; Section 6.1 compares the two.

5 Evaluation

5.1 Realism and diversity

Following GenIAS [26], each window is mapped to a fixed feature $\phi \in \mathbb{R}^{12}$. The window is z-scored, and ϕ holds the mean and standard deviation of the z-scored window (zero and one by construction; they are part of the frozen feature pack), its minimum and maximum, the mean and standard deviation of its first difference, its peak absolute value, the longest run of bins with absolute value above one as a fraction of W, and the log mean energy in four equal bands of

its Fourier spectrum. The z-scoring makes ϕ blind to a window's level and scale, so it measures shape. Nothing in ϕ is trained, and it is unrelated to the steering encoder.

Let A be the real anomalous windows of a held-out fold and G a generated gallery. With $\mathrm{dist}(a) = \min_{g \in G} \|\phi(a) - \phi(g)\|_2$,

$$\mathrm{ARP} = \frac{1}{1 + |A|^{-1} \sum_{a \in A} \mathrm{dist}(a)}. \tag{19}$$

ARP is high when every real anomaly lies close to some generated window. It rewards a gallery for reaching the real anomalies and ignores the rest of the gallery and the requested kind. Coverage, a fixed-radius analogue of the coverage of Naeem et al. [19], is the fraction of $a \in A$ with $\mathrm{dist}(a) \leq \rho$. The radius $\rho = 0.105$ is frozen: it is the value at which an unguided gallery sampled from pure noise covers 10% of the fold-0 anomalies.

Diversity has two summaries. The mean pairwise distance in ϕ is scale-dependent and unbounded. The entropy diversity index partitions the union of the real query windows and every gallery in the comparison by k-means with 16 clusters and a fixed seed, and reports the Shannon entropy of each gallery's share of those clusters, with ceiling $\log 16 \approx 2.77$. The partition depends on which galleries are included, so the entropies in Table 3 are comparable with each other only. We refit the partition on each fold and average.

Two classifier two-sample tests [16] ask whether a generated gallery can be told from the real anomalies. The first is a logistic regression on the raw window in channel-span units, balanced across channels, with a 70/30 split grouped by event, five seeds and a permutation test; it reports the median balanced accuracy. The second is a ridge classifier on standardized ROCKET features of the raw window, 300 random kernels fit on the donors [6], scored by out-of-fold AUC with folds grouped by event and averaged over three seeds. A value of 1/2 means chance on either scale.

Three galleries anchor the scale of ARP. Real anomalies from the training folds, offered as a gallery, score 0.792. The nominal donors score 0.320, and white noise scores 0.158. A gallery that stays close to its donors lands near 0.32 whatever its windows look like.

5.2 Detector utility

Detector. The detector is a small convolutional network: the convolution stack C of the steering encoder, trained separately here, followed by global average pooling and a linear map to one logit, 31,169 parameters in all. Inputs are in channel-span units. Training runs for 400 Adam steps at learning rate 10^{-3} with binary cross-entropy. Each batch holds 16 positives and 16 negatives drawn uniformly with replacement.

Data. For held-out fold k, the donors whose index modulo 3 equals k are test negatives: these 512 windows set the decision threshold, the empirical 1% false-alarm point. The other donors, less a fifth reserved as a reference set for rank fusion, are training negatives, together with the rare windows of the training folds. Positives are real anomaly windows of the training folds plus generated windows from that fold's gallery. A generated window whose donor is a test negative is excluded, so no test negative has an edited copy among the positives.

Scoring. An event of the held-out fold is found when any of its windows scores at or above the threshold. Event recall pools the three folds, 22 events, for each seed, and we report the median over five seeds. The false-alarm rate on rare-but-valid events is measured on the held-out fold's rare windows.

Budgets. At the full budget the detector sees every training-fold anomaly window, 396 to 546 of them, plus 256 generated windows. At budget k it sees k real windows, drawn uniformly once

per fold and seed and shared by every method, plus k generated windows. Differences between methods are therefore paired.

Intervals. For each method and event, the hit is averaged over seeds. The paired difference from real data alone is the mean over events of the difference in hits. Resampling the 22 events 2,000 times [8] gives a 95% percentile interval and a one-sided tail probability. With 22 events the intervals are wide, and we call a difference resolved only when its interval excludes zero.

Published baselines. Published and hand-built editors are given the same donors: a locked GenIAS checkpoint, a reimplementation of the published GenIAS procedure, CutAddPaste [24], and a hand-built injector. CutAddPaste pastes a patch of 8 to 35% of the window from another donor on the same channel, with a linear trend and Gaussian jitter. The injector adds a step, a pulse of 4 to 63 bins or a ramp, of 0.5 to 2 window standard deviations with a random sign, or rescales the window by $1 \pm u/4$ with u between 0.5 and 2. The reimplemented GenIAS differs from its donor by an RMSE of $2.3 \cdot 10^{-3}$ channel spans.

5.3 Galleries compared

Five galleries share the denoiser, the donors, the kind allocation and the seeds.

- The *contrast model* is the method of Section 4.
- The *masked* variant restores the RePaint mask: bins outside a mask around the requested anomaly follow the noised donor, and edits stay inside the mask.
- The *twin* runs the same chain with every edit off: a partly noised donor, denoised.
- The *prototype* gallery replaces the contrast on spikes and level shifts with a pull, in the steering encoder, toward the embedding of one real anomaly of the assigned kind from the other folds. The pull is a normalized gradient step of weight one, and the shell term and sampler are unchanged. The subsequence pool keeps the shell only.
- The *original-sampler prototype* uses the same targets on the sampler this work started from: a single causal pass without the reversed blend, the guidance gradient taken through the denoiser with respect to x_t, constant weights, no self-recurrence, and a full-strength last edit.

The sweep in Section 6.3 also includes the contrast model with the prototype pull added on top, and the no-edit twin of the masked variant.

6 Results

6.1 What the windows look like

Figure 5 shows real and generated windows, and Table 2 measures them. The spike measurements take the bin of largest deviation from the window's median. Height is that deviation; a spike occupies one bin when no other bin within eight exceeds half of it; baseline texture is the mean absolute first difference outside twelve bins around the peak. Level-shift roughness is the mean absolute first difference over the window.

The contrast model plants spikes at the real height, 0.838 against 0.831, and in a single bin in 99% of draws. Its level shifts carry the real step: scanned with the same level-shift contrast that sets δ, the median generated amplitude is 0.85 against 0.81 for the six real windows, and the transitions are 1.21 times as rough as the real event. The masked variant overshoots the spike height (0.988), spreads about half of its spikes over more than one bin, and makes level shifts 1.53 times as rough. The mask is the only difference between the two, so it accounts for

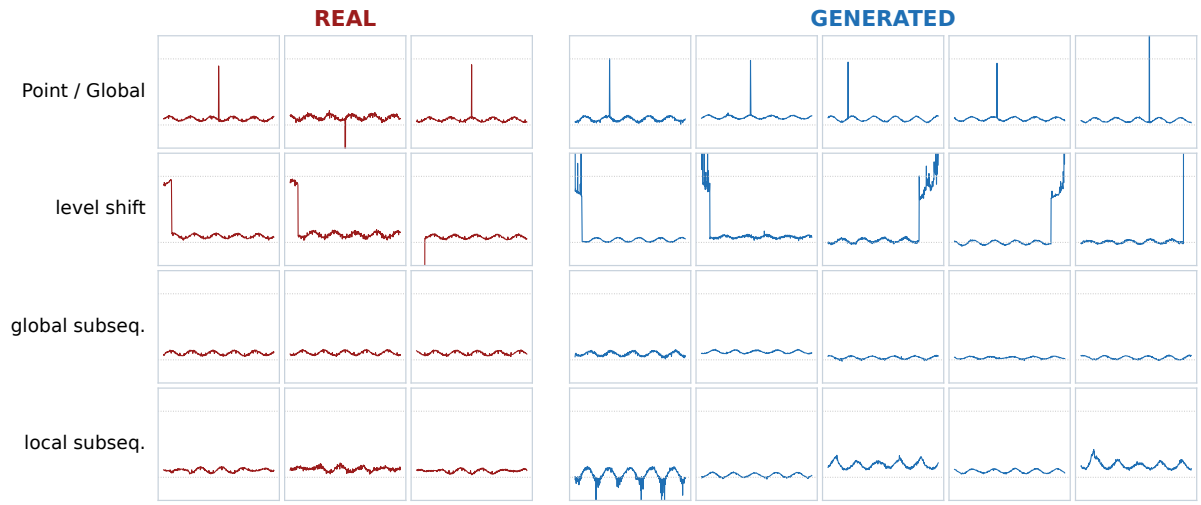


Figure 5: Three real windows and five generated windows of each kind, in channel-span units. Generated windows are random draws from the contrast model's three fold galleries; the subsequence rows show windows that received the global or local name. Dotted lines mark the channel span, 0 and 1.

Table 2: Morphology of point anomalies and level shifts, development galleries pooled over folds, channel-span units. Roughness and texture are ratios to the real windows. A noise burst is a point-anomaly draw whose baseline texture is more than three times the real one. Deep dips counts point-anomaly draws with three or more large downward excursions beside the requested spike. Real windows of short events are cut with the event in the middle, which fixes the real spike position at 0.50. Prototype is the prototype pull on our sampler; Orig. is the original-sampler prototype.

	Real	Contrast	Masked	Prototype	Orig.
Spike height	0.831	0.838	0.988	0.496	0.447
Single-bin spikes	100%	99%	53%	85%	53%
Spike position, 10th–90th pct.	0.50–0.50	0.18–0.82	0.18–0.82	0.50–0.50	0.04–0.50
Noise bursts	0%	3.0%	0%	21.8%	22.8%
Deep dips (≥ 3)	0%	14.8%	2.0%	—	48.4%
Baseline texture	1	0.83×	0.97×	1.08×	2.00×
Level-shift roughness	1	1.21×	1.53×	1.58×	1.70×
Level shift at the head	100%	50%	50%	100%	100%

the wider spikes and the rougher steps. It forces a seam between edited and donor bins, and the chain has to resolve that seam inside the mask. Both prototype galleries plant spikes at about half the real height, and about 22% of their point-anomaly draws are noise bursts.

Position separates the two families. The prototype on our sampler reproduces the real centring exactly (0.50–0.50), and every level shift it makes sits at the head of the window, where the one real event is. The contrast model places spikes where the sampled weight vector asks, between 0.18 and 0.82 of the window for 80% of draws, and puts half of its level shifts at the tail.

The contrast model has two visible defects. Three percent of its point draws are noise bursts, and 14.8% carry three or more deep downward dips in the baseline, which the real windows never show. The mask suppresses the dips (2.0%) by holding the baseline at the donor; the original-sampler prototype has them in 48% of draws. Around its spikes the contrast model's baseline is slightly smoother than the data, at 0.83 times the real texture.

The edit is spread over the window. Relative to its twin, the contrast model changes 42% of all bins by more than 0.05 channel spans, and 29% of the bins of a point-anomaly window. The masked variant changes 0.9% of the bins of a point-anomaly window, essentially the spike, so its

Table 3: Realism on the development folds. ARP, coverage and pairwise distance are means over the three held-out folds. Entropy is the mean of three partitions, each fit to that fold's real anomalies together with every gallery listed here. The ROCKET two-sample AUC was computed for the sampler variants only. The real row has no ARP, which would be the distance from a set to itself.

Gallery	ARP	Cov.	Entropy	Pairwise	ROCKET AUC
Real anomalies	—	—	1.06	—	—
Contrast (ours)	0.509	0.038	2.33	10.53	0.880
Contrast, masked	0.499	0.033	2.32	9.68	0.881
Prototype	0.579	0.029	2.37	8.83	0.882
Prototype, original sampler	0.567	0.015	2.34	7.30	0.966
Twin (no edit)	0.318	0.081	0.97	1.71	0.905
GenIAS, locked	0.317	0.195	0.72	1.09	—
GenIAS, reimplemented	0.290	0.053	0.17	0.80	—
CutAddPaste	0.322	0.290	0.72	1.11	—
Hand-built shapes	0.319	0.267	0.74	1.21	—
Nominal donor	0.320	0.354	0.71	1.11	—

mask doubles as a bin-level label.

6.2 Feature-space realism

Table 3 is the GenIAS protocol. The contrast model reaches ARP 0.509, against 0.318 for its twin and about 0.32 for every published or hand-built editor, which stay at the donor. The GenIAS reimplementation is slightly below the donor, consistent with its small RMSE. On the anchor scale, 0.509 lies between the donor (0.320) and an oracle gallery of real anomalies (0.792). The prototype galleries score higher, 0.579 on our sampler and 0.567 on the original one. Per-fold ARP intervals from an event bootstrap are wide, about 0.2 to 0.8, and the four steered galleries lie well inside each other's intervals; the paired comparison with the twin is the informative one.

That paired gain averages +0.191, interval $[+0.039, +0.346]$, from fold-wise gains of +0.143, +0.051 and +0.381. Standardizing ϕ shrinks the three to +0.025, +0.005 and +0.202. Coverage moves the other way, from 8.1% for the twin to 3.8% for the contrast model: the average distance falls, and so does the fraction of real anomalies inside the radius. The misses are mostly global subsequences, which dominate A and which the pool reaches through the shell alone. Entropy is 2.33 for the contrast gallery and 1.06 for the real anomalies in the same partition, so the gallery spreads into clusters the real set leaves empty. Pairwise distance is 10.5, against 1.7 for the twin and about 1.1 for the editors.

The two classifier tests read separability. On raw windows, the logistic regression tells the twin from real anomalies at a median balanced accuracy of 0.517, close to chance, and the steered galleries at 0.63 to 0.66 (contrast 0.645, masked 0.634, original-sampler prototype 0.662). ROCKET features separate every gallery well, with AUC 0.905 for the twin, 0.880 for the contrast model, 0.882 for the prototype and 0.966 for the original-sampler prototype. By the first test steering makes a window easier to tell from real anomalies; by the second the contrast model and the prototype are slightly harder to tell apart than the twin. The sampler optimizes neither test.

6.3 Event recall

At the full budget, median event recall is 0.591 for real anomalies alone and 0.682 with the contrast gallery added (mean 0.655, average precision 0.485, window AUROC 0.617). The paired difference is +0.091, interval $[-0.045, +0.236]$, so it is unresolved. Against GenIAS and the hand-built shapes the paired differences are +0.118 $[-0.036, +0.273]$ and +0.109 $[-0.018, +0.245]$. GenIAS, its reimplementation and CutAddPaste all reach a median of 0.545, and GenIAS minus

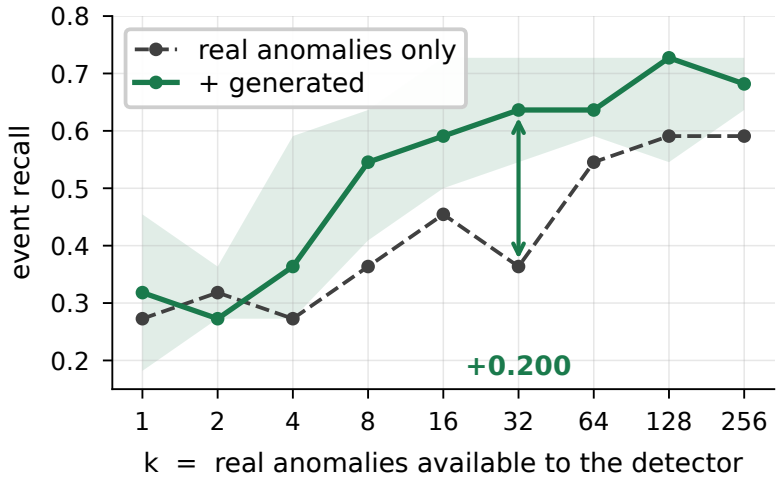


Figure 6: Event recall of a detector trained on k real anomaly windows (dashed) or on those windows plus k windows from the contrast model (solid), median over five seeds. The shaded band is the range of the solid curve over the seeds. The annotation at $k = 32$ is the paired difference +0.200; the two medians differ by more, because the pairing is per event.

real data alone is -0.027, interval $[-0.055, 0]$. The hand-built shapes reach 0.591. The contrast model raises the false-alarm rate on rare-but-valid events from 0.026 to 0.059, so part of the recall is paid for in alarms on operations that are unusual and valid.

The prototype gallery reaches a full-budget median of 0.727, with a rare false-alarm rate of 0.088, and the original-sampler prototype reaches 0.636. At this budget the ordering follows ARP more closely than spike morphology: a window-level network can use a large deviation of any shape. The twin reaches 0.500 in a second run of the same protocol, in which real data alone again reaches 0.591, so regenerated donors without an edit lower recall when they are added as positives.

Figure 6 and Table 4 are the comparison the generator is for: k real anomaly windows plus k generated ones. For the contrast model the interval excludes zero at $k = 8$, 32 and 64, with gains of +0.155, +0.200 and +0.136; at $k = 32$ the interval is $[+0.073, +0.318]$. At $k \leq 4$ the generated windows add nothing measurable, and from $k = 128$ on the gain is about +0.11 with intervals that include zero. The original-sampler prototype has the largest gains up to $k = 32$ and is resolved from $k = 2$ to $k = 32$; from $k = 64$ on, the contrast model's point estimates exceed it. The prototype on our sampler is resolved at $k = 32$ only. Adding the prototype pull on top of the contrast gives +0.182 at $k = 32$, and restoring the mask gives +0.109. The no-edit twin of the masked variant lowers recall: its paired difference is negative at seven of the nine budgets, and its interval lies below zero at $k = 2$, 4, 64 and 256.

Rank-fusing the detector logit with a label-free diffusion reconstruction score changes the full-budget comparison: the paired difference between the contrast model and real data alone becomes +0.136, with interval $[+0.036, +0.264]$. Fusion makes it a different system, and we keep the detector alone as the primary endpoint.

6.4 Held-out test

The 91 test events have not been used. The pre-registered test report is one run of the contrast model, with the frozen ϕ and the same detector, at the full budget and at $k = 32$, with no retuning after the numbers are seen. Until that run, every claim above is a claim about the 22 development events.

Table 4: Event recall at budget k: median over five seeds, and paired change Δ relative to the same k real windows alone, bootstrapped over 22 events. Medians can tie while paired differences differ, because the pairing is per event. Prototype is the prototype pull on our sampler; Orig. is the original-sampler prototype.

	Real	Contrast		Prototype		Orig.	
k	median	median	Δ	median	Δ	median	Δ
1	0.273	0.318	+0.009	0.318	+0.036	0.409	+0.091
2	0.318	0.273	−0.009	0.318	−0.018	0.500	+0.136*
4	0.273	0.364	+0.027	0.364	+0.082	0.545	+0.182*
8	0.364	0.545	+0.155*	0.500	+0.100	0.545	+0.182*
16	0.455	0.591	+0.164	0.545	+0.118	0.636	+0.200*
32	0.364	0.636	+0.200*	0.591	+0.164*	0.636	+0.236*
64	0.545	0.636	+0.136*	0.636	+0.100	0.636	+0.127
128	0.591	0.727	+0.118	0.682	+0.127	0.636	+0.091
256	0.591	0.682	+0.109	0.727	+0.118	0.636	+0.082

*95% bootstrap interval excludes zero. Two intervals end exactly at zero and are left unmarked: Prototype at $k = 8$ and Orig. at $k = 64$.

7 Limitations

Sample size. Every interval in this paper is computed on 22 events, and the level-shift numbers describe one event. The three fold-wise ARP gains, +0.143, +0.051 and +0.381, show how much a single fold can move a mean.

Subsequences. Global and local subsequences are steered by the shell alone. On this split the shell energy ranks labelled anomalies against nominal windows at an AUROC of about 0.54, so Q means unusual among nominal windows and carries no information about labelled subsequences. Those two kinds are 648 of the 672 real windows. A budget sweep that draws real windows uniformly is mostly a sweep over subsequences; a sweep stratified by kind is future work.

Labels inside the encoder. The steering encoder is the fold-0 checkpoint on every fold, and its contrastive loss was trained on the anomalies of folds 1 and 2. For those two folds the shell gradient is taken in a representation that has seen their labels. The contrast term is independent of the encoder; the subsequence pool depends on it entirely. The level shift has a related exception: all six windows are one event in fold 0, so the other folds borrow its amplitude and fold 0 measures the amplitude on its own held-out event.

Gallery and windows. Spikes and level shifts are each a quarter of a gallery, and the subsequence pool is the other half, split into three names of 256. In the real windows, 636 of 672 are global subsequences. The twin uses the same donors and the same balance and lowers recall, so the balance alone does not produce the gain, but the gallery still departs from the mission's prior. The held-out windows are cut like the training windows, with short events in the middle, so the event recall endpoint gives no credit for the position spread of Table 2. The prototype galleries win ARP and full-budget recall with spikes at half the real height; the contrast model plants the requested shape. Bin-level localisation, where a known position would matter, is left open, and the masked variant is the only recipe that emits a bin-level label.

Published baselines. Beating GenIAS, CutAddPaste and the hand-built shapes is a weak claim on these donors: their ARP equals the donor's to within 0.03, and the reimplemented GenIAS stays within an RMSE of $2.3 \cdot 10^{-3}$ of its donor. The comparisons that carry weight are the twin, the two prototype galleries, and real data alone.

Model scope. The denoiser is causal apart from (3), the schedule is linear and ends at $\bar{\alpha}_{199} = 0.13$, and every sample starts from a donor at $t_0 = 40$. Other channels and other missions are untouched. A different backbone or schedule would leave the subsequence target and the 22-event intervals as they are.

8 Conclusion

A nominal diffusion model can be steered into spikes and level shifts whose size matches the few labelled examples, using a shell quantile on every window and an exact linear contrast on the two kinds that have one. On the development events of ESA-ADB Mission 1, channels 41–46, the generated windows help a small detector when real anomalies number in the tens, and the gain is unresolved once every development anomaly is available. Prototype pulls toward real anomaly embeddings lead on feature-space realism, on full-budget recall and at the smallest budgets, with spikes at half the real height and positions copied from the labelled windows. The contrast model plants the requested shape at a sampled position, and about one point-anomaly draw in seven still carries a run of deep baseline dips. We stop at this recipe. Subsequence targets, a bin-level detector, a kind-stratified budget and one pre-registered run on the sealed test are the open measurements. A bidirectional backbone, a different noise schedule and sampling from pure noise are future work.

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